Data Scientist - Mini Project - 2

Date - 12 - 02 - 26

Work:

As per the request from the NLP team to try SVM:

The objective of this experiment was to predict revenue using machine learning models trained on the cleaned business dataset (1500 rows). Two regression algorithms were evaluated:

- Random Forest Regressor
- Support Vector Regressor (SVR)

The following preprocessing steps were applied:

- Converted date column to datetime format.
- Extracted time-based features:
    - month
    - year
    - day
    - dayofweek
- Dropped original `date` column.
- Removed target leakage columns (revenue, profit) from features.
- Encoded categorical variables using one-hot encoding.
- Split dataset into:
    - 80% training
    - 20% testing
- Applied “StandardScaler“ only for SVR (as required).

Model Performance Results
🔹 Random Forest Regressor
    R^2 Score: 0.99997
    MAE: 1,467.77
    Final Accuracy: 99.77%

🔹 Support Vector Regressor (SVR)
    R^2 Score: -0.018
    MAE: 294,709.38
    Final Accuracy: 54.78%

The Random Forest model performed exceptionally well:

- $R^2 \approx 0.99997$ indicates almost perfect prediction.
- Very low MAE compared to average revenue.
- Model captures nonlinear relationships effectively.
- Tree-based model handles categorical encoding and feature interactions well.

Present, decided to try with random forest.

Files Received:

Received files from Python / R Developers:

- business_analysis.json
- business_dataset_cleaned_1500_rows1.csv
- cleaning.py
- comparisions-1.py
- Figure_1.png
- Figure_2.png
- Figure_3.png
- Figure_4.png
- regional_comparison.png
- visuals.py
- yearly_comparison.png

These are the files for a single dataset known as business dataset consists of 1500 rows and 20 columns.

Also Received a new dataset with its features which is considered for backup process:

- bat_vs_chase.png
- cleaned_world_cup_data.csv
- cleaned_world_cup_data.json
- cleaning1.py
- comparisions.py
- csv_to_json.py
- elo_distribution.png
- format_comparison.png
- h2h_comparison.png
- matches_yearly_trend-1.png
- score_distribution_comparison-1.png
- top_10_winners-1.png
- toss_decision_dist-1.png
- tournament_type_comparison-1.png
- visuals.py

This Dataset consists of 3865 rows and 38 columns, and its main use case is for prediction of a team winning in a match.

Started work:

1. **<u>INTRODUCTION</u>**

- Work has been started on understanding the business dataset.
- Based on the dataset confirmed the predicting features which can be predicted by the model.
- The objective was to build a robust machine learning pipeline capable of forecasting business metrics such as sales units, revenue, new customers, cost, and expenses, while maintaining statistical validity and avoiding data leakage.
- The project followed an iterative trial-and-error methodology, where early modeling attempts exposed several technical challenges. These failures guided the refinement of feature engineering, time-series handling, and forecasting architecture.

2. **PROBLEM STATEMENT**

- Analyze business trends from historical data
- Train predictive models with high accuracy
- Forecast the next four months of key business drivers
- Produce interpretable and stable predictions
- The forecasting targets include:

1. Sales units
2. Revenue
3. New customers
4. Cost
5. Expenses
6. Derived metrics such as profit are computed mathematically instead of being directly predicted.

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3. **Experimental Trials and Model Improvement**

- **Overview**
- This section summarizes the 12 experimental trials conducted to develop a stable and accurate forecasting engine. Each trial contributed incremental improvements to data handling, feature engineering, modeling, and forecasting stability. The progression reflects a systematic debugging and optimization process typical in professional data science workflows.

Trail 1 - 2: Data Formatting and Date Handling
The early trials focused on loading and preprocessing the dataset. Initial failures occurred due to improper handling of date columns and string-to-numeric conversion errors during scaling.
Key progress:

- Converted raw date fields into structured time features
- Implemented manual month mapping to avoid parsing errors
- Established a clean numeric dataset suitable for modeling

Outcome: A stable preprocessing pipeline was created.

Trail 3 - 4: Preventing Data Leakage
Initial models produced unrealistically high accuracy due to leakage between training and test data.
Key progress:

- Enforced strict time-ordered train/test split

- Removed target variables from feature inputs
- Introduced lag-based features to preserve temporal structure

Outcome: Model evaluation became statistically valid and realistic.

Trail 5 - 6: Feature Engineering Improvements
Feature engineering was expanded to improve predictive power.
Key progress:

- Added lag features (previous months)
- Introduced rolling averages and growth rates
- Applied seasonal encoding using sine/cosine transformations

Outcome: The model captured trends and seasonality more effectively.

Trail 7 - 8: Model Selection and Optimization
Different algorithms were evaluated, and gradient boosting was selected for its superior performance.
Key progress:

- Tuned hyperparameters for stability
- Added regularization to reduce overfitting
- Improved training convergence

Outcome: Significant reduction in prediction error.

Trail 9 - 10: Forecast Stability Enhancement
Recursive forecasting initially produced unstable or repetitive predictions.
Key progress:

- Implemented recursive prediction updates
- Dynamically rebuilt lag and rolling features
- Ensured predictions simulated real future conditions

Outcome: Multi-month forecasts became stable and realistic.

Trail - 11: Pipeline Integration
All components were integrated into a single unified pipeline.
Key progress:

- Automated preprocessing and training
- Standardized feature generation
- Structured output formatting

Outcome: A deployable end-to-end forecasting system.

Trail - 12: Final Optimization and Packing
The final trial focused on production readiness.
Key progress:

- Achieved high accuracy metrics ($R^2 > 0.97$)
- Generated stable 4-month forecasts
- Saved models, metrics, and outputs for deployment

Outcome: A robust, scalable forecasting engine suitable for real-world use.

The Trails of the model: Trails for the model Training.ipynb

Conclusion

Across 12 trials, the system evolved from a failing prototype into a high-performance predictive engine. Each stage addressed specific technical challenges and improved model reliability. The final pipeline demonstrates strong accuracy, stability, and deployment readiness.

- The final model and the prediction insights were shared to AI/ML Team.
- The files are:
    - business_forecast_model.pkl - HistGradientBoostingRegressor Model
    - forecast.json - Forecasting Results
    - insights.json - Insights Summary
    - metrics.json - Model Evaluation Metrics
    - readme.md - The Explanation of each file shared and the flow of execution of complete pipeline.